Universidad Zaragoza

Automatic Machine Learning

Applications of Deep Learning (69158)

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Machine Learning

"Machine learning is the science of getting computers to act without being explicitly programmed."

by Andrew Ng

Zoubin Ghahramani said that he often heard that:
"I'd like to use machine learning, but I can't invest much time."

Goal of AutoML

AutoML

The goal of AutoML is to automate all parts of machine learning
(as needed) to *support* users efficiently building their ML-applications.

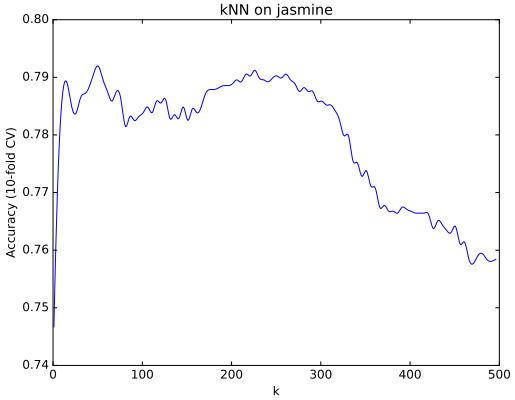
Informal Definition: AutoML System

Given

- a dataset
- a task (e.g., regression or classification)
- a cost metric (e.g., accuracy or RMSE)

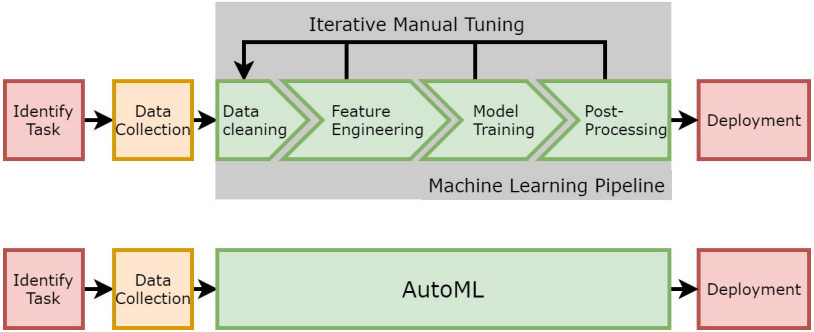
an AutoML system automatically determines the approach that performs best for this particular application.

A Simple Example with k-NN



- k-nearest neighbors is one of the simplest ML algorithms
- Size of neighbourhood (k) is very important for its performance
- The performance function depending on k is quite complex (not at all convex)

ML vs AutoML



With AutoML, we ...

- support ML users
- improve the efficiency of developing new ML applications
- reduce the required ML-expertise
- might achieve better performance than developers w/o AutoML

Challenges in AutoML

1. Each dataset potentially requires different optimal ML-designs
   - Design decisions have to be made for each dataset again
2. Training of a single ML model can be quite expensive (e.g., hours, days or weeks)
   - often, we cannot try many design decisions
3. the mathematical relation between design decisions and performance is (often) unknown
   - gradient-based optimization is not directly possible
4. optimization in highly complex spaces
   - incl. categorical choices, continuous parameters, conditional dependencies

Building a Successful AutoML Tool

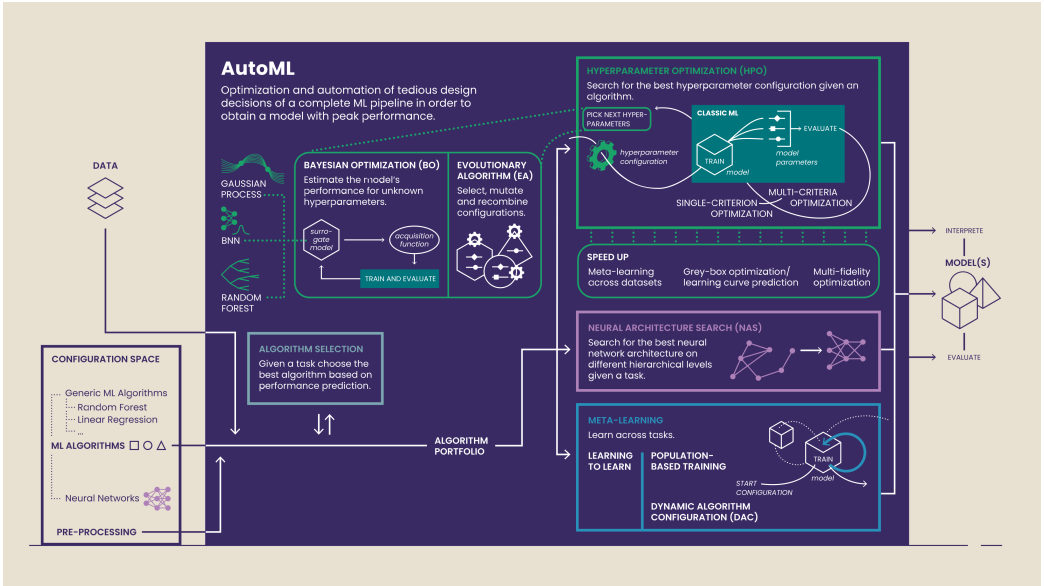
$\rightsquigarrow$ There is not a single way for building successful AutoML tools, but there are well established ideas.

- **HPO** Search for the best hyperparameter configuration of a ML algorithm
- **CASH** Search for the best combination of algorithm and hyperparameter configuration
- **NAS** Search for the architecture of neural network

- **Selection** Predict the best algorithm (and its hyperparameter configuration)
- **DAC** Predict the best hyperparameter configuration for an algorithm state at a given time point

$\rightsquigarrow$ But how can we combine all of these?

AutoML overview



Model Parameters vs. Hyperparameters

Model parameters are optimized during training, typically via loss minimization. They are an **output** of the training.

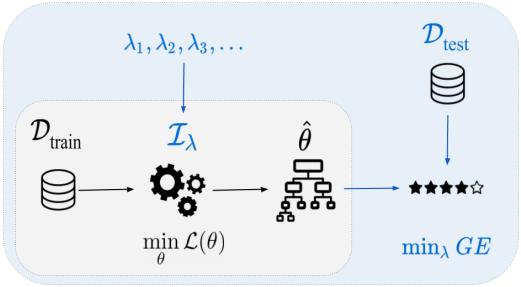
- Example: Network weights and biases.

Hyperparameters (HPs) must be specified *before* the training, they are an **input** of the training. Hyperparameters often control the complexity of a model or any computational part of the training process.

- Examples: Number of layers, learning rate, etc.

Hyperparameter Tuning: A bi-level optimization problem I

We face a **bi-level** optimization problem:



The inner loop optimizes the model parameters θ using the loss function $\mathcal{L}$ (e.g.: MSE, cross-entropy, etc.). The outer loop optimizes the hyperparameters λ for the performance metric (e.g.: generalization error (GE)).

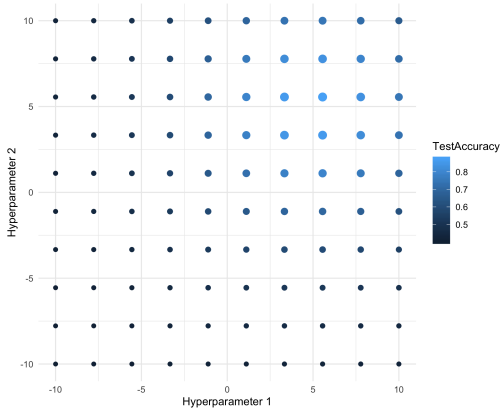
Hyperparameter Tuning: A bi-level optimization problem II

The components of a tuning problem are:

- The dataset (warning: at least two level splits *train-validation-test*)
- The learner (possibly: several competing learners?) that is tuned
- The learner's hyperparameters and their respective regions-of-interest over which we optimize
- The performance measure, as determined by the application.
  Not necessarily identical to the loss function that defines the loss minimization problem for the learner!

Grid search I

- Simple technique which is still quite popular, tries all HP combinations on a multi-dimensional discretized grid
- For each hyperparameter a finite set of candidates is predefined
- Then, we simply search all possible combinations in arbitrary order



Grid search over 10x10 points

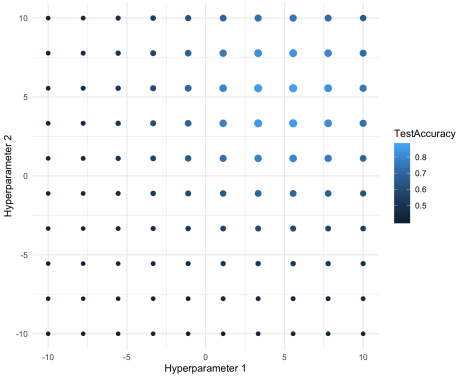
Grid search II

Advantages

- Very easy to implement
- All parameter types possible
- Parallelizing computation is trivial

Disadvantages

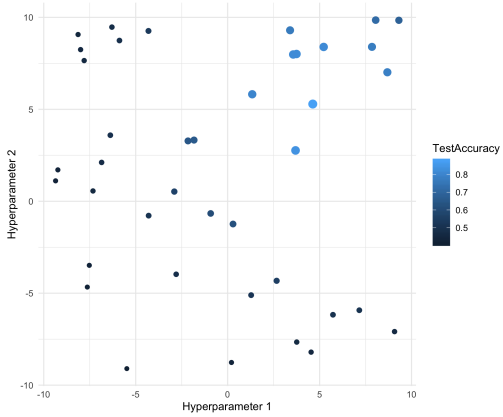
- Scales badly: Combinatorial explosion
- Inefficient: Searches large irrelevant areas
- Low resolution in each dimension
- Arbitrary: Which values / discretization?



Grid search over 10x10 points

Random search I

- Small variation of grid search
- Uniformly sample from the region-of-interest



Random search over 100 points

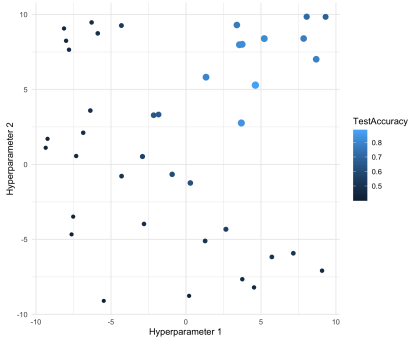
Random search II

Advantages

- Very easy to implement, all parameter types possible, trivial parallelization
- Anytime algorithm: more budget $\Rightarrow$ better performance.
- No discretization required

Disadvantages

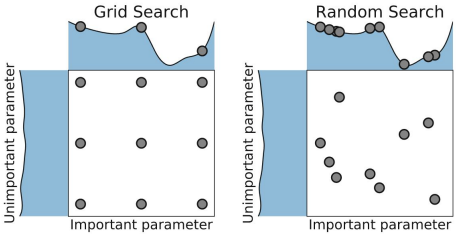
- Inefficient: many evaluations in areas with low likelihood for improvement
- Scales badly: high dimensional hyperparameter spaces need *lots* of samples to cover.



Random search over 100 points

Grid search vs. Random search

- With a tuning budget of T only $T^{\frac{1}{d}}$ unique hyperparmeter values are explored for each $\lambda_1, \ldots, \lambda_d$ in a grid search.
- Random search will (most likely) see T different values for each hyperparameter.
- Grid search can be disadvantageous if some hyperparameters have little or no influence on the model.



Comparison of grid search and random search. [Hutter et al. 2019]

Actually...


SIGOPT
GRID SEARCH
WWW.NOGRIDSEARCH.COM
SIGOPT
GRID SEARCH
WWW.NOGRIDSEARCH.COM
SIGOPT
RANDOM SEARCH
WWW.NORANDOMSEARCH.COM
SIGOPT
GRID SEARCH
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Global Blackbox Optimization

- Consider the global optimization problem of finding:

$$\boldsymbol{\lambda}^* \in \arg\min_{\boldsymbol{\lambda} \in \boldsymbol{\Lambda}} f(\boldsymbol{\lambda})$$

- In the most general form, function f is a blackbox function:

  - Only mode of interaction with f: querying f's value at a given $\boldsymbol{\lambda}$
  - Function f may not be available in closed form, not convex, not differentiable, noisy, etc.

- Hyperparameter optimization (HPO) can be dfined as a blackbox optimization problem
  - Define $f(\boldsymbol{\lambda}) := \mathcal{L}_{HPO}(\mathcal{A}_{\boldsymbol{\lambda}}, \mathcal{D}_{train}, \mathcal{D}_{valid})$

Bayesian Optimization: Origin of the Name

- Bayesian optimization uses Bayes' theorem:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)} \propto P(B|A) \times P(A)$$

- Bayesian optimization uses this to compute a posterior over functions:

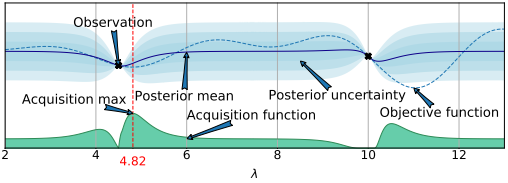
$$P(f|\mathcal{D}_{1:t}) \propto P(\mathcal{D}_{1:t}|f) \times P(f), \qquad \text{where } \mathcal{D}_{1:t} = \{\boldsymbol{\lambda}_{1:t}, c(\boldsymbol{\lambda}_{1:t})\}$$

- Meaning of the individual terms:
  - $P(f)$ is the prior over functions, which represents our belief about the space of possible objective functions before we see any data
  - $\mathcal{D}_{1:t}$ is the data (or observations, evidence)
  - $P(\mathcal{D}_{1:t}|f)$ is the likelihood of the data given a function
  - $P(f|\mathcal{D}_{1:t})$ is the posterior probability over functions given the data

Bayesian Optimization of a blackbox function in a nutshell

General approach

- Fit a probabilistic model to the collected function samples $\langle \boldsymbol{\lambda}, c(\boldsymbol{\lambda}) \rangle$
- Use the model to guide optimization, trading off exploration *vs* exploitation



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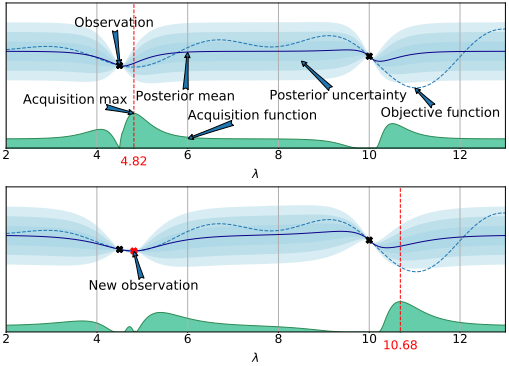

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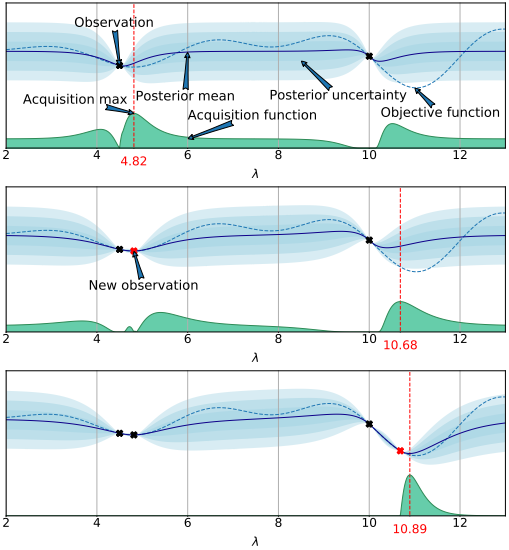


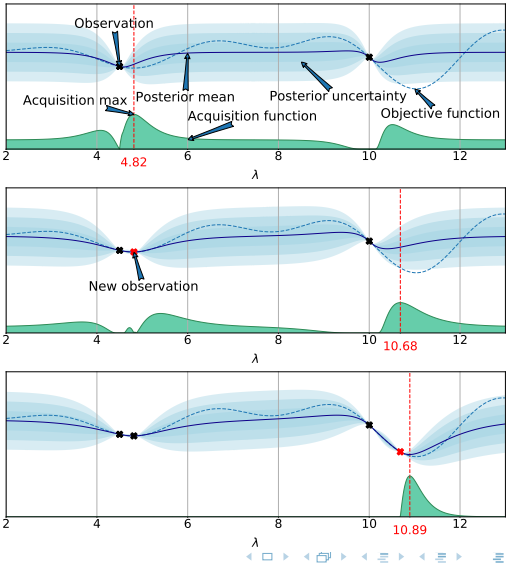
Bayesian Optimization of a blackbox function in a nutshell

General approach

- Fit a probabilistic model to the collected function samples $\langle \boldsymbol{\lambda}, c(\boldsymbol{\lambda}) \rangle$
- Use the model to guide optimization, trading off exploration *vs* exploitation

Popular approach in the statistics literature since [Mockus. 1975]

- Efficient in #function evaluations
- Works when objective is nonconvex, noisy, has unknown derivatives, etc.
- Recent convergence results [Srinivas et al. 2009; Bull. 2011; de Freitas et al. 2012; Kawaguchi et al. 2015]



Bayesian Optimization: Advantages and Disadvantages

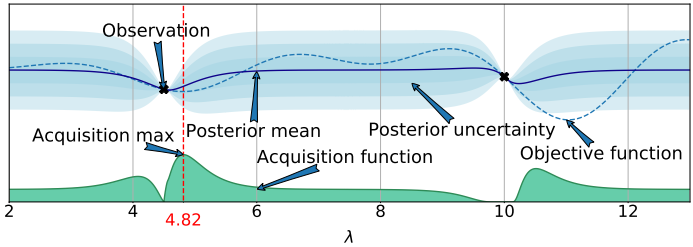
Advantages

- Sample efficient
- Can handle noise
- Native incorporation of priors
- Does not require gradients
- Theoretical guarantees

Disadvantages

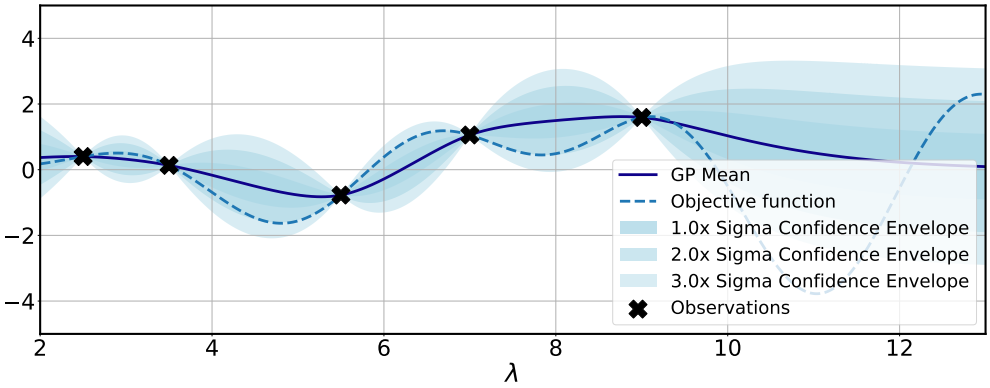
- Overhead because of model training in each iteration
- Crucially relies on robust surrogate model

Acquisition Functions: the Basics



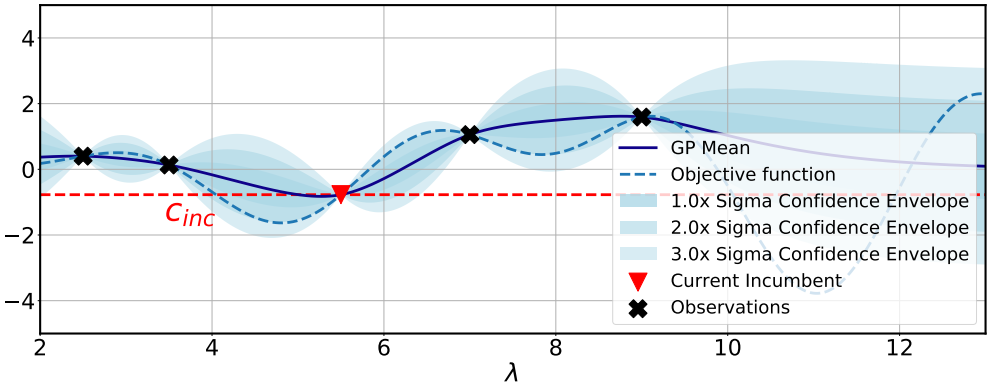
- Given the surrogate model $\hat{c}^{(t)}$ at the t-th iteration of BO, the acquisition function $u(\cdot)$ judges the utility (or usefulness) of evaluating f at $\boldsymbol{\lambda}^{(t)} \in \boldsymbol{\Lambda}$ next
- The acquisition function needs to trade off exploration and exploitation
  - E.g., just picking the $\boldsymbol{\lambda}$ with lowest predicted mean would be too greedy
  - We also need to take into account the uncertainty of the surrogate model $\hat{c}^{(t)}$ to explore

Probability of Improvement (PI): Concept



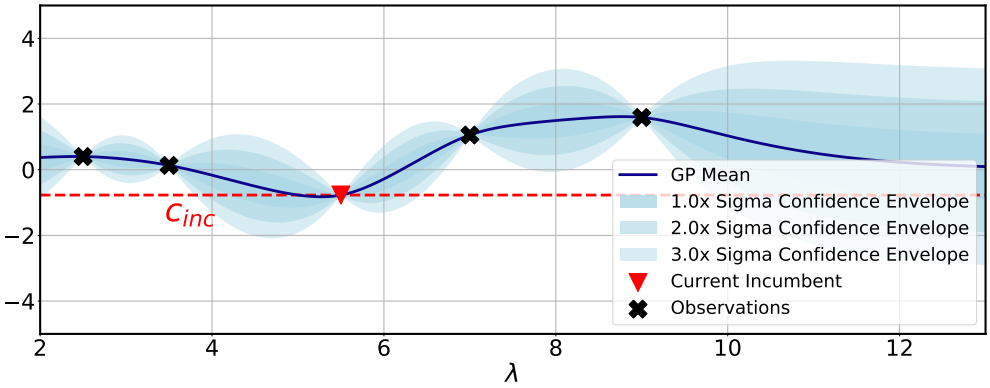
Given the surrogate fit at iteration t

Probability of Improvement (PI): Concept



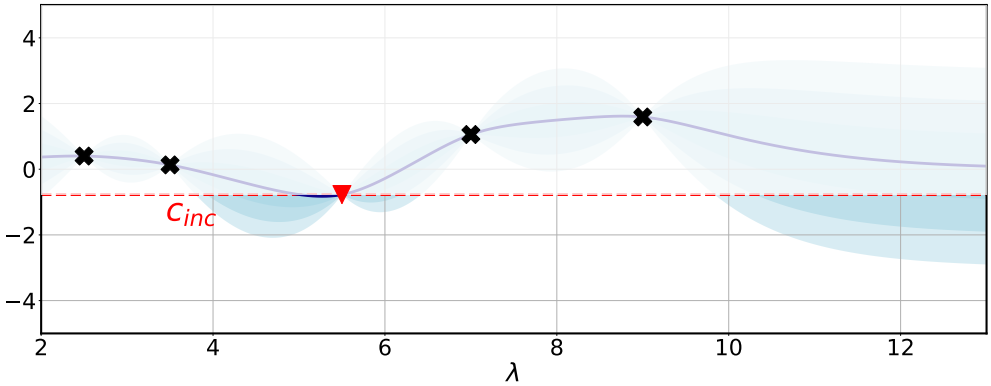
Current incumbent $\hat{\boldsymbol{\lambda}}$ and its observed cost c_{inc}

Probability of Improvement (PI): Concept



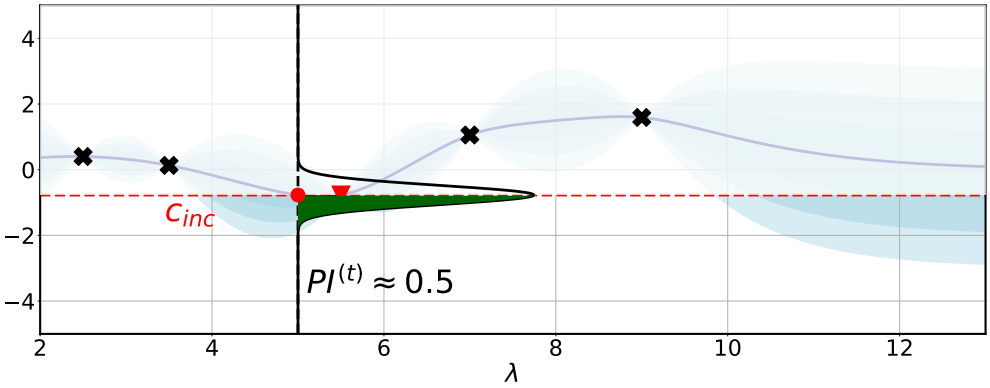
Now let's drop the objective function - it's unknown after all!

Probability of Improvement (PI): Concept



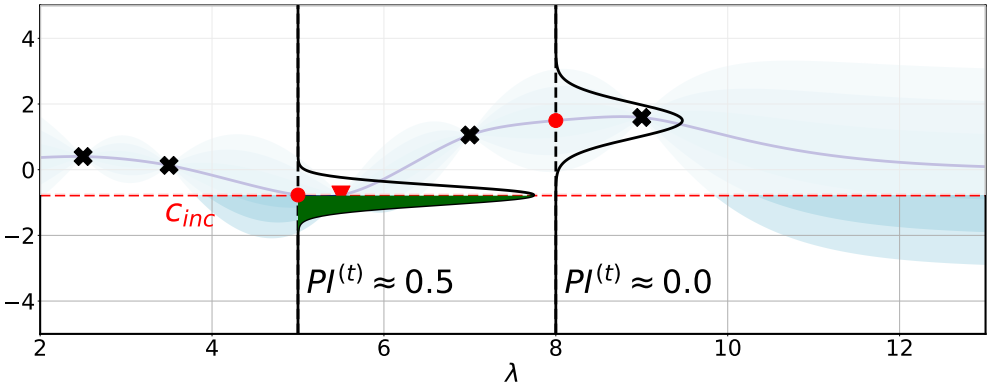
Intuitively, we care about the probability of improving over the current incumbent

Probability of Improvement (PI): Concept



PDF of a good candidate configuration. Only the green area is an improvement.

Probability of Improvement (PI): Concept



PDF of a bad candidate configuration

Probability of Improvement (PI): Formal Definition

- We define the current incumbent at time step t as: $\hat{\boldsymbol{\lambda}}^{(t-1)} \in \arg\min_{\boldsymbol{\lambda}' \in \mathcal{D}^{(t-1)}} c(\boldsymbol{\lambda}')$
- We write c_{inc} shorthand for the cost of the current incumbent: $c_{inc} = c(\hat{\boldsymbol{\lambda}}^{(t-1)})$
- The probability of improvement $u_{PI}(\boldsymbol{\lambda})$ at a configuration $\boldsymbol{\lambda}$ is then defined as:

$$u_{PI}^{(t)}(\boldsymbol{\lambda}) = P(c(\boldsymbol{\lambda}) \leq c_{inc}).$$

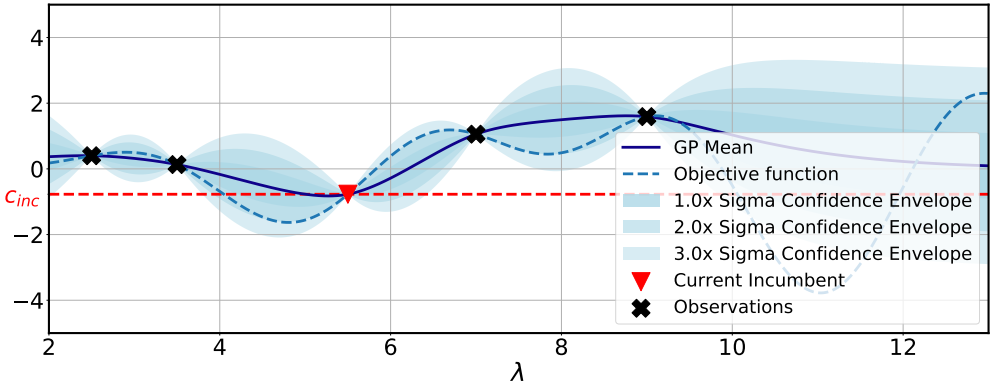
- Since the predictive distribution for $c(\boldsymbol{\lambda})$ is a Gaussian $\mathcal{N}(\mu^{(t-1)}(\boldsymbol{\lambda}), \sigma^{2(t-1)}(\boldsymbol{\lambda}))$, this can be written as:

$$u_{PI}^{(t)}(\boldsymbol{\lambda}) = \Phi[Z], \quad \text{with } Z = \frac{c_{inc} - \mu^{(t-1)}(\boldsymbol{\lambda}) - \xi}{\sigma^{(t-1)}(\boldsymbol{\lambda})},$$

where $\Phi(\cdot)$ is the CDF of the standard normal distribution and ξ is an optional exploration parameter

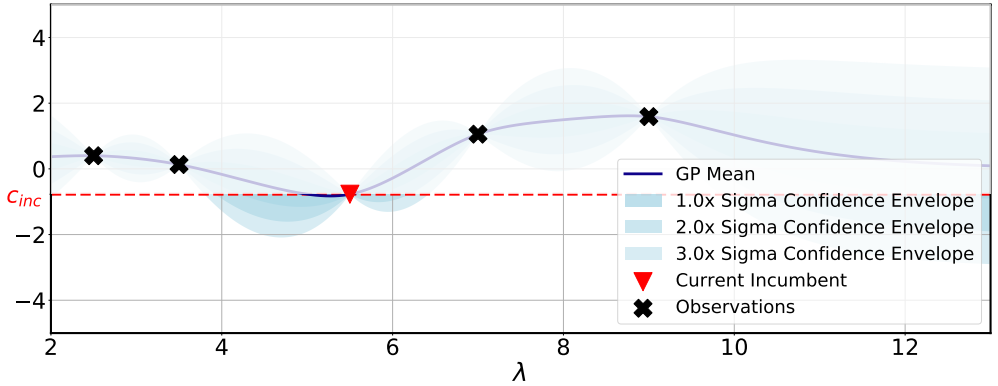
$$\boxed{\text{Choose } \boldsymbol{\lambda}^{(t)} \in \arg\max_{\boldsymbol{\lambda} \in \boldsymbol{\Lambda}} (u_{PI}^{(t)}(\boldsymbol{\lambda}))}$$

Expected Improvement (EI): Concept



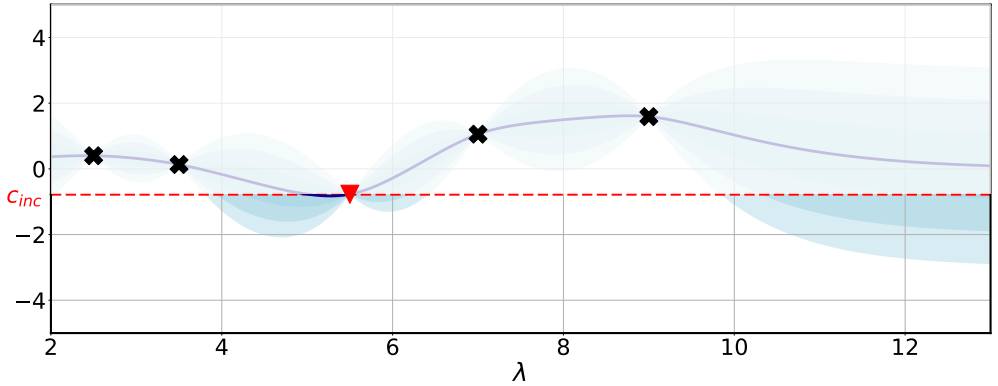
Given the surrogate fit at iteration t

Expected Improvement (EI): Concept



Region of probable improvement – but how large is the improvement?

Expected Improvement (EI): Concept



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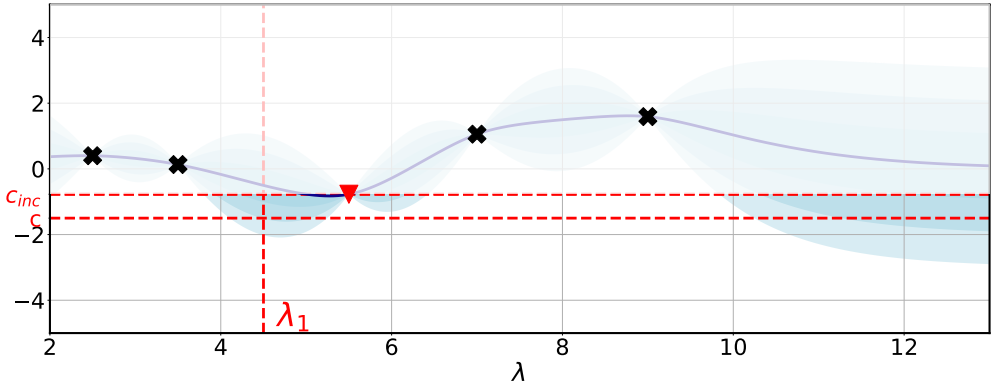
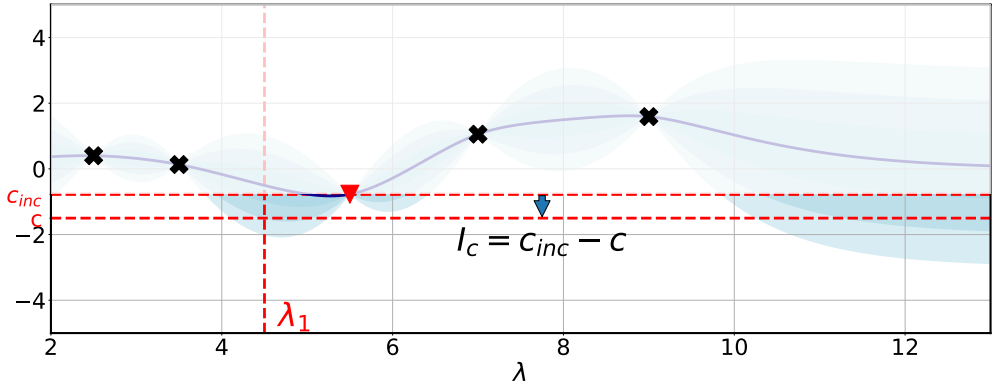


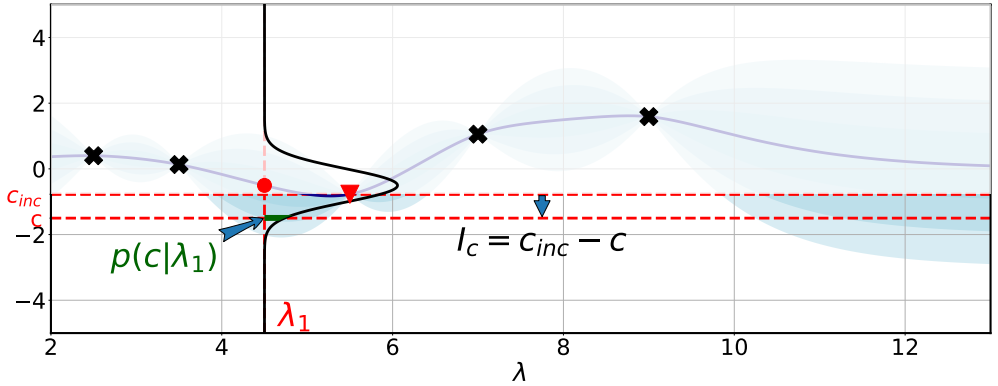
Hypothetical *real* cost c at a given $\boldsymbol{\lambda}$ - unknown in practice without evaluating

Expected Improvement (EI): Concept



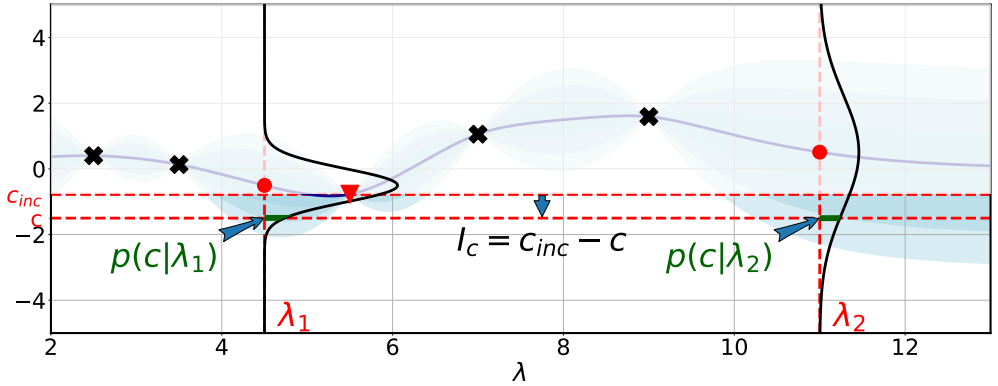
Given a hypothetical c, we can compute the improvement $I_c(\boldsymbol{\lambda})$

Expected Improvement (EI): Concept



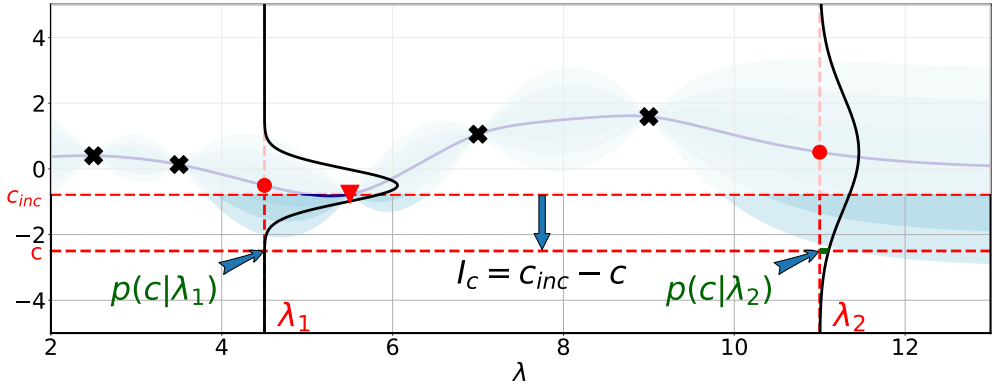
Given $\hat{c}(\boldsymbol{\lambda}) = \mathcal{N}(\mu(\boldsymbol{\lambda}), \sigma^2(\boldsymbol{\lambda}))$, we can also compute $p(c|\boldsymbol{\lambda})$.

Expected Improvement (EI): Concept



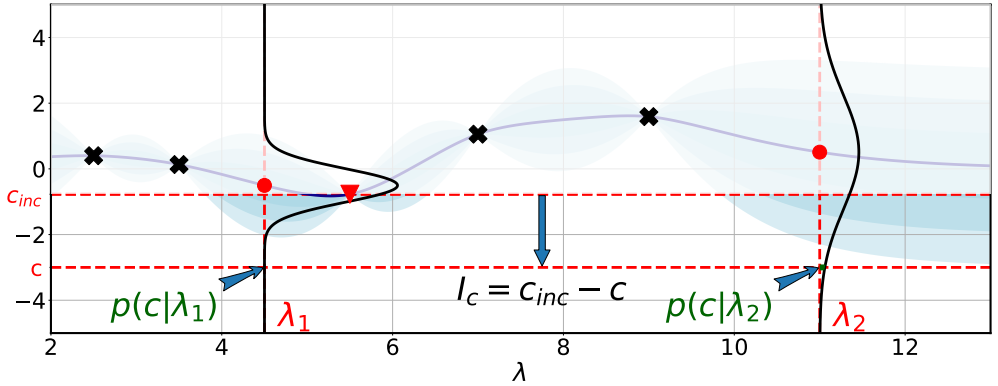
Compare the likelihood of a given improvement for two different configurations $\boldsymbol{\lambda}_1$ and $\boldsymbol{\lambda}_2$

Expected Improvement (EI): Concept



Now consider the likelihood of a larger improvement.

Expected Improvement (EI): Concept



Larger improvements are more likely in areas of high uncertainty.

To compute $\mathbb{E}[I(\boldsymbol{\lambda})]$, intuitively, we sum $p(c \mid \boldsymbol{\lambda}) \times I_c$ over all possibles values of c.

Expected Improvement (EI): Formal Definition

- We define the one-step positive improvement over the current incumbent as

$$I^{(t)}(\boldsymbol{\lambda}) = \max(0, c_{inc} - c(\boldsymbol{\lambda}))$$

- Expected Improvement is then defined as

$$u_{EI}^{(t)}(\boldsymbol{\lambda}) = \mathbb{E}[I^{(t)}(\boldsymbol{\lambda})] = \int_{-\infty}^{\infty} p^{(t)}(c \mid \boldsymbol{\lambda}) \times I^{(t)}(\boldsymbol{\lambda}) \; dc.$$

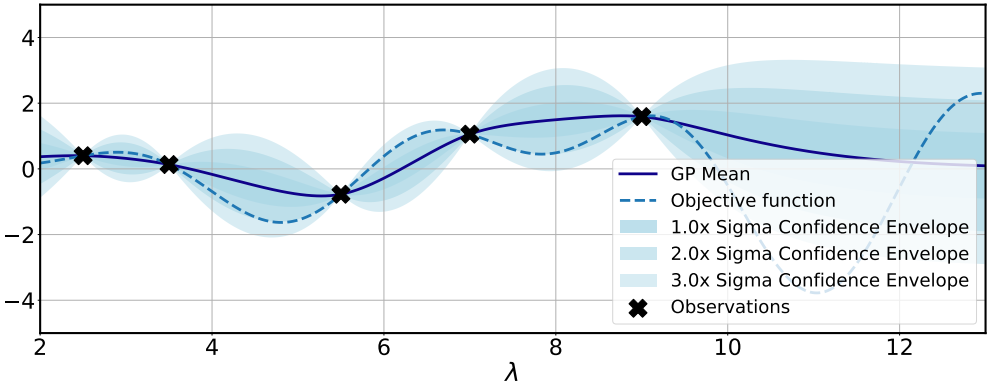
- Since the posterior distribution of $\hat{c}(\boldsymbol{\lambda})$ is a Gaussian, EI can be computed in closed form:

$$u_{EI}^{(t)}(\boldsymbol{\lambda}) = \begin{cases} \sigma^{(t)}(\boldsymbol{\lambda})[Z\Phi(Z) + \phi(Z)], & \text{if } \sigma^{(t)}(\boldsymbol{\lambda}) > 0 \\ 0 & \text{if } \sigma^{(t)}(\boldsymbol{\lambda}) = 0, \end{cases}$$

$$\text{where } Z = \frac{c_{inc} - \mu^{(t)}(\boldsymbol{\lambda})}{\sigma^{(t)}(\boldsymbol{\lambda})}$$

$$\boxed{\text{Choose } \boldsymbol{\lambda}^{(t)} \in \arg\max_{\boldsymbol{\lambda} \in \boldsymbol{\Lambda}} (u_{EI}^{(t)}(\boldsymbol{\lambda}))}$$

Lower/Upper Confidence Bounds (LCB/UCB): Concept



Given the surrogate fit at iteration t

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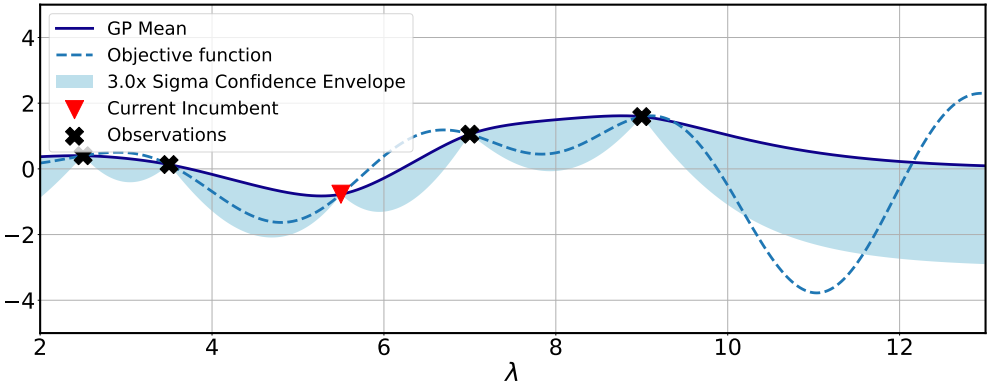

Lower Confidence Bound, $\mu(\boldsymbol{\lambda}) - \alpha\sigma(\boldsymbol{\lambda})$ (here, for $\alpha = 3$)

Lower/Upper Confidence Bounds (LCB/UCB): Formal Definition

- We define the Lower Confidence Bound as

$$u_{LCB}^{(t)}(\boldsymbol{\lambda}) = \mu^{(t)}(\boldsymbol{\lambda}) - \alpha\sigma^{(t)}(\boldsymbol{\lambda}), \quad \alpha \geq 0$$

- One can schedule α (e.g., increase it over time [Srinivas et al. 2009])

$$\boxed{\text{Choose } \boldsymbol{\lambda}^{(t)} \in \arg\max_{\boldsymbol{\lambda} \in \boldsymbol{\Lambda}} \left(-u_{LCB}^{(t)}(\boldsymbol{\lambda})\right)}$$

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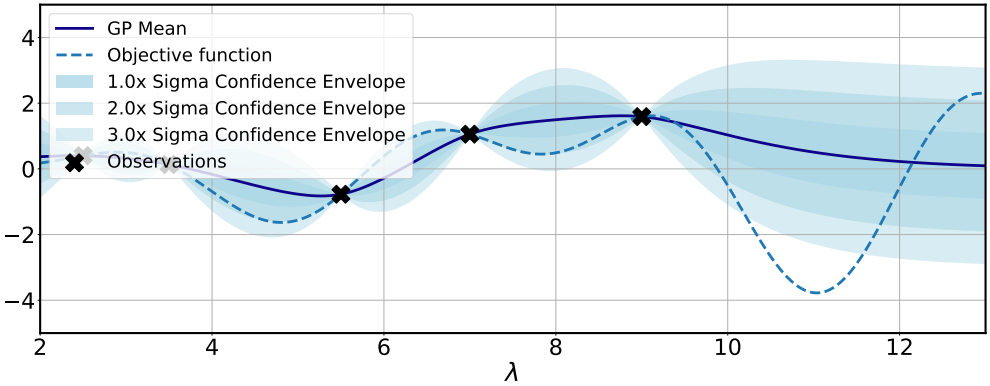
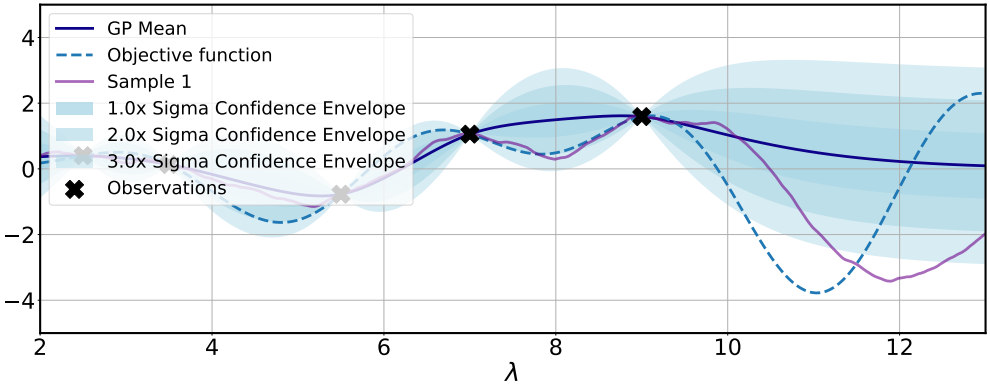
- Note: when one aims to maximize the objective function, one would use UCB instead
  - $u_{UCB}^{(t)}(\boldsymbol{\lambda})) = \mu^{(t)}(\boldsymbol{\lambda}) + \alpha\sigma^{(t)}(\boldsymbol{\lambda})$
  - For UCB, one would choose $\boldsymbol{\lambda}^{(t)} \in \arg\max_{\boldsymbol{\lambda} \in \boldsymbol{\Lambda}}(u_{UCB}^{(t)}(\boldsymbol{\lambda}))$

Thompson Sampling (TS): Concept



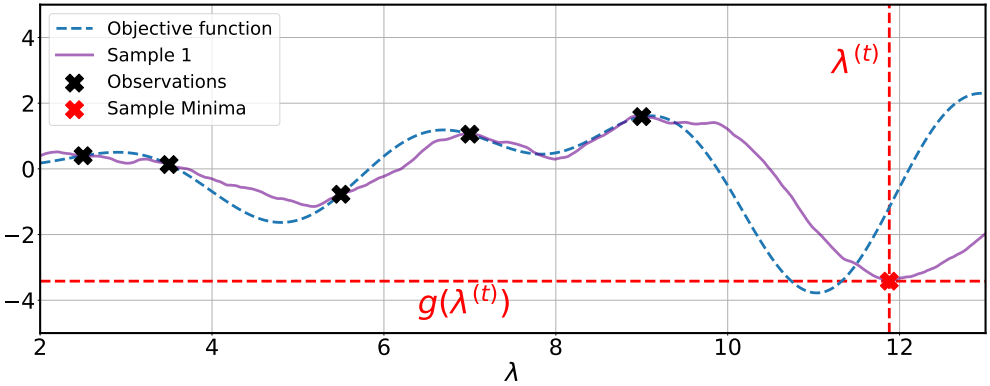
Given the surrogate at iteration t fit on dataset $\mathcal{D}^{(t-1)}$

Thompson Sampling (TS): Concept



Draw a sample g from the predictive surrogate model

Thompson Sampling (TS): Concept



Then choose the minimum of this sample to evaluate at next

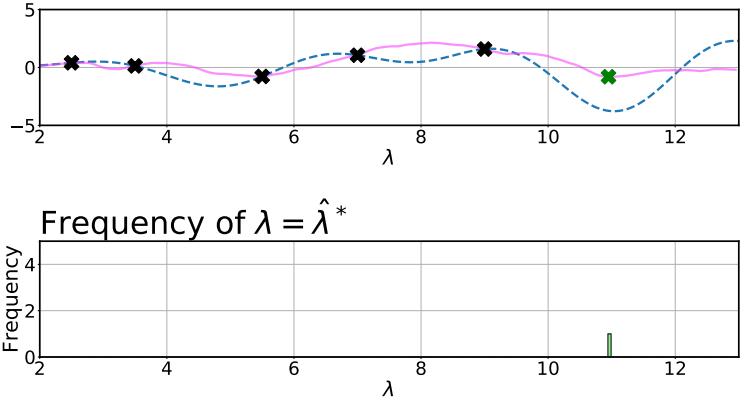
Entropy Search Preliminaries

- Key idea: Evaluate $\boldsymbol{\lambda}$ which most reduces our uncertainty about the location of $\boldsymbol{\lambda}^*$
- We'll use the p_{min} distribution to characterize the location of $\boldsymbol{\lambda}^*$:

$$p_{min}(\boldsymbol{\lambda}|\mathcal{D}) = p(\boldsymbol{\lambda} \in \arg\min_{\boldsymbol{\lambda}' \in \boldsymbol{\Lambda}}(\hat{c}(\boldsymbol{\lambda}')|\mathcal{D}))$$

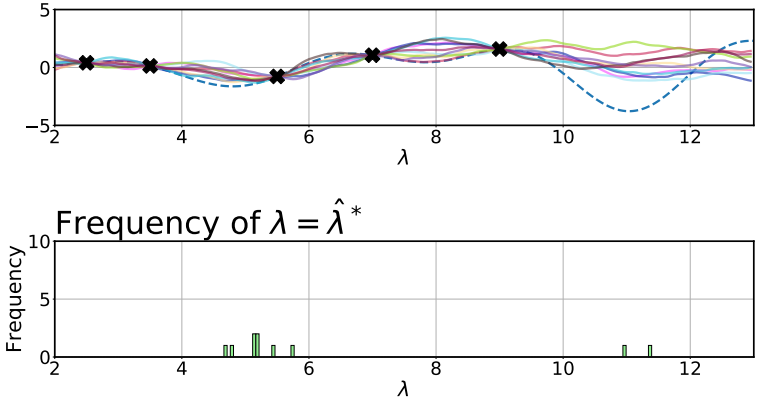
- Our uncertainty is then captured by the entropy $H(p_{min}(\cdot|\mathcal{D}))$ of the p_{min} distribution
- Minimizing $H(p_{min}(\cdot|\mathcal{D}))$ yields a peaked p_{min} distribution, i.e., strong knowledge about the location of $\boldsymbol{\lambda}^*$

Entropy Search: Visualization of the p_{min} Distribution



For each sample drawn from $\hat{c}$, we can compute where $\boldsymbol{\lambda}^*$ lies

Entropy Search: Visualization of the p_{min} Distribution



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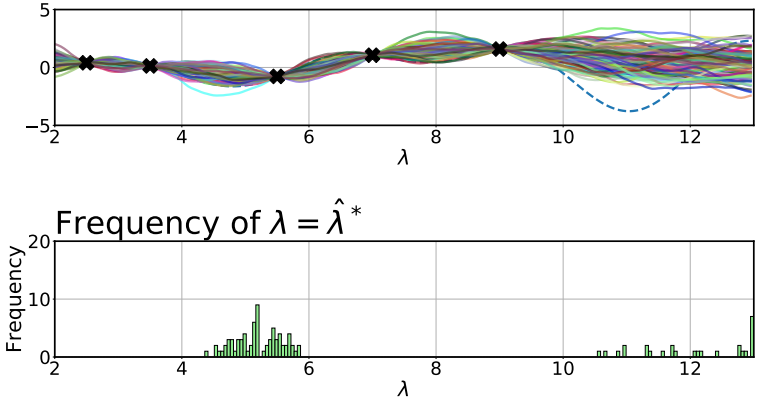
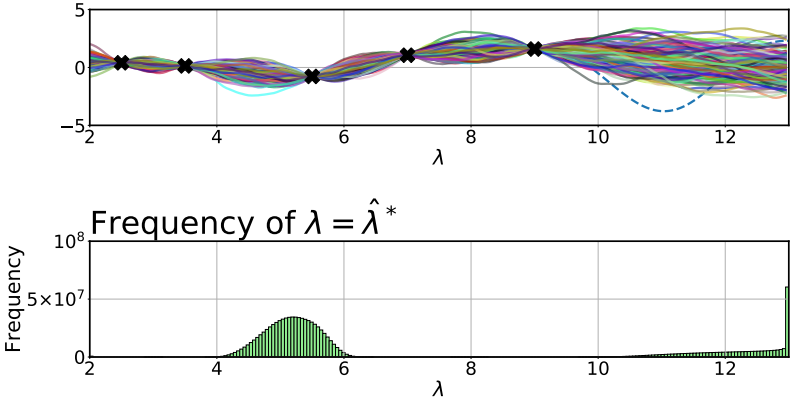


From many samples we can approximate the p_{min} distribution

Entropy Search: Visualization of the p_{min} Distribution



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Entropy Search: Formal Definition

- The p_{min} distribution characterizes the location of $\boldsymbol{\lambda}^*$:

$$p_{min}(\boldsymbol{\lambda}^*|\mathcal{D}) = p(\boldsymbol{\lambda}^* \in \arg\min_{\boldsymbol{\lambda}' \in \boldsymbol{\Lambda}}(\hat{c}(\boldsymbol{\lambda}')|\mathcal{D}))$$

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- Entropy search aims to minimize $H(p_{min})$, to yield a peaked p_{min} distribution:

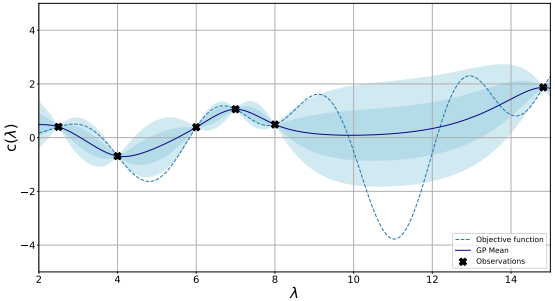
$$u_{ES}(\boldsymbol{\lambda}) = H(p_{min}(\cdot|\mathcal{D})) - \mathop{\mathbb{E}}_{\tilde{c} \sim \hat{c}(\lambda)^{(t)}} H(p_{min}(\cdot|\mathcal{D} \cup \{\langle \boldsymbol{\lambda}, \tilde{c} \rangle\}))$$

$$\text{Choose } \boldsymbol{\lambda}^{(t)} = \arg\max_{\boldsymbol{\lambda} \in \boldsymbol{\Lambda}}(u_{ES}^{(t)}(\boldsymbol{\lambda}))$$

Desiderata for Surrogate Models in Bayesian Optimization

In all cases

- Regression model with uncertainty estimates
- Accurate predictions



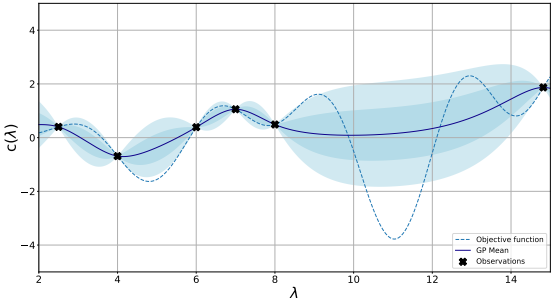
Desiderata for Surrogate Models in Bayesian Optimization

In all cases

- Regression model with uncertainty estimates
- Accurate predictions

Examples:

- Gaussian Processes
- Bayesian Neural Networks
- Random Forests



Gaussian Processes (GPs): Reminder of Pros and Cons

Advantages

- Smooth and reliable uncertainty estimates
- Strong sample efficiency
- We can encode expert knowledge about the design space in the kernel

These advantages make GPs the most commonly-used model in Bayesian optimization

Disadvantages

- Performance can be quite sensitive to the choice of kernel
- Cost scales cubically with the number of observations
- Weak performance for high dimensionality
- Not easily applicable in discrete or conditional spaces
- Sensitive to its own hyperparameters

Bayesian Neural Networks (BNNs): Overview of Pros and Cons

Advantages

- Scales linearly with #observations
- Can obtain nice and smooth uncertainty estimates
- Flexibility: handling of categorical, continuous and discrete spaces

Disadvantages

- Usually needs more data than Gaussian processes
- Uncertainty estimates often worse than for Gaussian processes
- Many meta-design decisions
- No robust off-the-shelf implementation

These qualities make BNNs an ever-more promising alternative

Beyond the Standard Bayesian Optimization Setting

Standard Bayesian optimization problems

- Low-dimensional functions
- Continuous, smooth functions
- Sequential optimization

Extensions

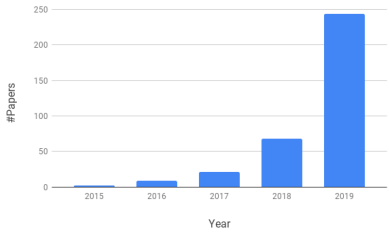
- Structured search spaces: categorical & conditional hyperparameters
- High dimensions
- Parallel evaluations
- Optimization with constraints

Neural Architecture Search (NAS)

- Goal: automatically find neural architectures with strong performance
  - Optionally, subject to a resource constraint

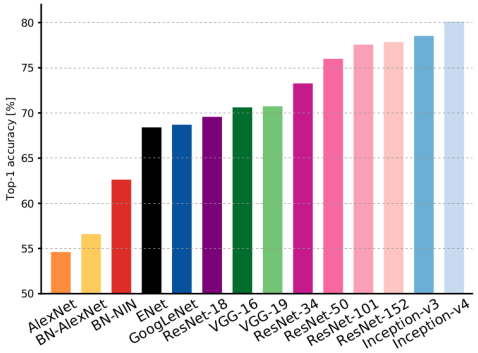
- A decade-old problem, but main stream since 2017 and now intensely researched
- One of the main problems AutoML is known for

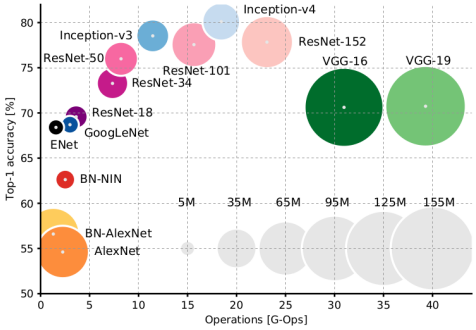
- Initially extremely expensive
- By now several methods promise low overhead over a single model training



Motivation for NAS

- Performance improvements on various tasks due to novel architectures
- Can we automate this design process, potentially discovering new components/topologies?

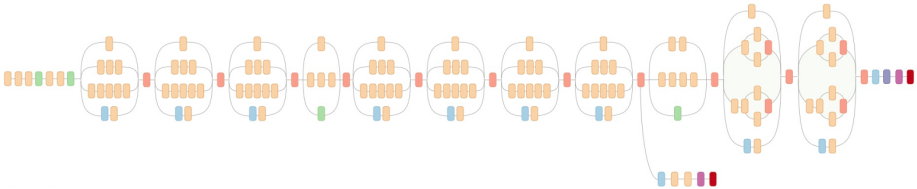




[Canziani et al. 2017]

Motivation for NAS

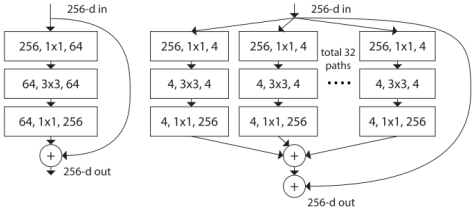
- Manual design of architectures is time consuming
- Complex state-of-the-art architectures are a result of years of trial and errors by experts



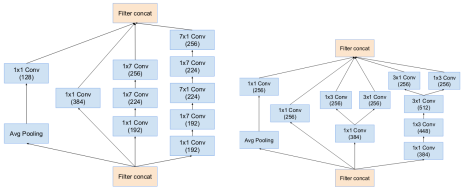
Inception-v3 [Szegedy et al. 2015]

Motivation for NAS

- Manual design of architectures is time consuming
- Complex state-of-the-art architectures are a result of years of trial and errors by experts
  - Main pattern: Repeated blocks with same structure (topology)

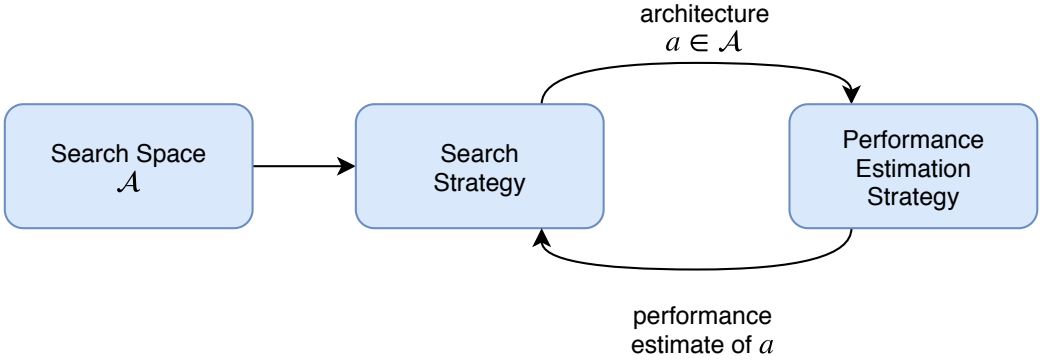


ResNet/ResNeXt blocks
[He et al. 2016; Xie et al. 2016]



Inception-v4 blocks [Szegedy et al. 2016]

NAS components [Elsken et al. 2019]



- Search Space: the types of architectures we consider; micro, macro, hierarchical, etc.
- Search Strategy: Reinforcement learning, evolutionary strategies, Bayesian optimization, gradient-based, etc.
- Performance Estimation Strategy: validation performance, lower fidelity estimates, one-shot model performance, etc.

Bibliography:

- This lecture is based on:
  - **AutoML course**, Marius Lindauer, Frank Hutter, Bernd Bischl, Lars Kotthoff, Janek Thomas `https://ki-campus.org/index.php/courses/automl-luh2021`
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